MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal choice.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A square is inscribed in a circle of radius r. The area of the square is: [1]

(A) r^2 (B) $2r^2$ (C) $4r^2$ (D) $\sqrt{2}r^2$

Q2. The area of the incircle of an equilateral triangle of side 42 cm is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 231 cm^2 (B) 462 cm^2 (C) 924 cm^2 (D) 154 cm^2

Q3. The perimeter of a sector of a circle of radius 7 cm with central angle 90° is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 11 cm (B) 22 cm (C) 25 cm (D) 35 cm

Q4. The area of a sector whose perimeter is four times its radius r units is: [1]

(A) $\frac{r^2}{4}$ sq. units (B) r^2 sq. units (C) $2r^2$ sq. units (D) $4r^2$ sq. units

Q5. Assertion (A): If the length of the minute hand of a clock is 7 cm, then the area swept by it in 5 minutes is $\frac{77}{6}$ cm^2.

Reason (R): The length of an arc of a sector of angle θ and radius r is given by $l = \frac{\theta}{360^\circ} \times 2\pi r$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. The minute hand of a clock is 10.5 cm long. Find the area described by the minute hand on the face of the clock between 7:00 am and 7:10 am. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. An athletic running track has two straight parallel segments of length 106 m each, and the distance between the two inner parallel lines is 60 m. The track is 10 m wide everywhere. Find: (i) the distance around the track along its inner edge, and (ii) the total area of the track. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

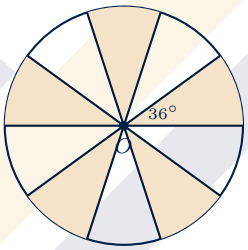
Q8. A round table cover has six equal designs. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of 0.35 per cm^2. $\left(\text{Use } \sqrt{3} = 1.7, \pi = \frac{22}{7}\right)$ [3]

Q9. On a square handkerchief of side 42 cm, nine circular designs each of radius 7 cm are printed in three rows of three circles each. Find the area of the remaining portion of the handkerchief. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Jeweller's Filigree Silver Brooch Architecture

A goldsmith crafts a silver filigree brooch in the form of a circle of diameter $d = 35$ mm. Thin silver wire is used for the circular rim and also to forge five diameters that divide the brooch into ten equal sectors.



(i) Find the central angle subtended by each individual sector of the brooch. [1]

(ii) Calculate the total length of the silver wire required to manufacture the brooch. [1]

(iii) Calculate the surface area of each sector of the brooch, OR calculate the total circular face area of the brooch. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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